1.8 Reaction Kinetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which of the following factors can affect the value of the activation energy of a reaction?

1 The presence of a catalyst

2 Changes in temperature

3 Changes in the concentration of the reactants

A. 1 only

B. 1 and 2

C. 3 only

D. 1, 2 and 3

[1 mark]

Question 2

The oxidation of butadiene, CH_2=CHCH=CH_2, using air or oxygen, produces the molecule crotonaldehyde, CH_3CH=CHCHO.

One method of oxidation is to pass a mixture of butadiene and oxygen through a hot aqueous solution of palladium(II) ions, Pd^{2+}(aq), which catalyse the reaction.

Which statement about the action of the Pd^{2+} (aq) ions is **not** correct?

A. Pd^{2+} (aq) lowers the activation energy for the reaction.

B. Pd^{2+} (aq) increases the energy of the reacting molecules.

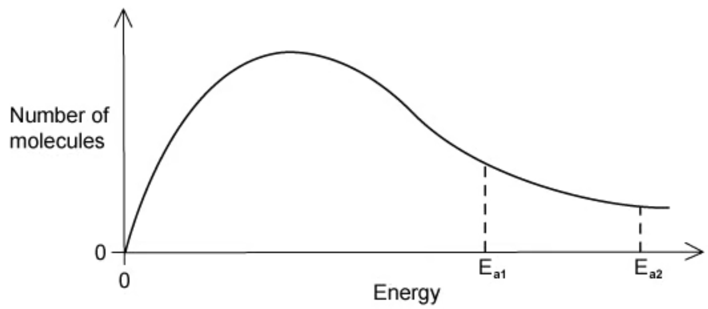
C. When Pd^{2+} (aq) is used, the reaction proceeds by a different route.

D. Changing the concentration of the Pd^{2+} (aq) affects the rate of oxidation.

[1 mark]

Question 3

The Maxwell-Boltzmann energy distribution curve below describes a mixture of two gases at a given temperature. For a reaction to occur between the gaseous molecules, they must collide with sufficient energy.



Of the two activation energy (E_a) values shown, one is for a catalysed reaction, the other for an uncatalysed one.

When a catalyst is used, which pair of statements is correct?

A	E_{a1} uncatalysed reaction fewer effective collisions	E_{a2} catalysed reaction more effective collisions
B	E_{a1} catalysed reaction fewer effective collisions	E_{a2} uncatalysed reaction more effective collisions
C	E_{a1} uncatalysed reaction more effective collisions	E_{a2} catalysed reaction fewer effective collisions
D	E_{a1} catalysed reaction more effective collisions	E_{a2} uncatalysed reaction fewer effective collisions

[1 mark]

Question 4

Ammonia is manufactured using the Haber process, which is represented by the following equation:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g) \quad \Delta H = -93\ \text{kJ mol}^{-1}$$

What happens to the rate of the forward and backward reactions when the temperature is increased?

A. There is no effect on the backward or forward rate.

B. Both forward and backward rates increase.

C. The forward rate only increases.

D. The backward rate only increases.

[1 mark]

Question 5

Why does a mixture of hydrogen gas and bromine gas react together faster at a temperature of 500 K than it does at a temperature of 400 K?

1 A higher proportion of effective collisions occurs at 500 K.

2 Hydrogen molecules and bromine molecules collide more frequently at 500 K.

3 The activation energy of the reaction is lower at 500 K.

A. 1 only

B. 1 and 2

C. 3 only

D. 1, 2 and 3

[1 mark]

Question 6

When the pressure of a fixed mass of gaseous reactants is raised at a constant temperature, the rate of reaction increases.

Which of the following statements explain this observation?

1 Raising the pressure lowers the activation energy.

2 More molecules have energy greater than the activation energy at the higher pressure.

3 More collisions occur per second when the pressure is increased.

A. 1 only

B. 1 and 2

C. 3 only

D. 1, 2 and 3

[1 mark]

Question 7

What is the main reason for the increase in reaction rate with increasing temperature?

A. The activation energy decreases.

B. The activation energy increases.

C. The molecules collide more frequently.

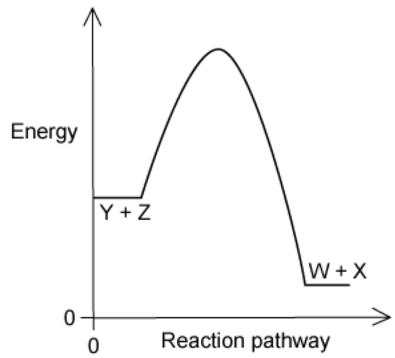
D. More molecules have an energy greater than the activation energy.

[1 mark]

Question 8

The diagram represents the reaction pathway for the following reaction:

$$Z(g) + Y(g) \rightarrow X(g) + W(g)$$



What statement is true about the reverse reaction, $W(g) + X(g) \rightarrow Y(g) + Z(g)$?

A. It will have a smaller activation energy and a negative ΔH

B. It will have a smaller activation energy and a positive ΔH

C. It will have a larger activation energy and a negative ΔH

D. It will have a larger activation energy and a positive ΔH

[1 mark]

Question 9

A student measured the rate of a reaction at two different temperatures: 40 °C and 50 °C. They observed that the rate of reaction roughly doubled.

What explains this observation?

A. Raising the temperature by 10 °C doubles the average velocity of the molecules.

B. Raising the temperature by 10 °C doubles the average kinetic energy of each molecule.

C. Raising the temperature by 10 °C doubles the number of molecules having more than a certain minimum energy.

D. Raising the temperature by 10 °C doubles the number of molecular collisions in a given time.

[1 mark]

Question 10

Which statements correctly describe how a catalyst works?

1 A catalyst has no effect on the enthalpy change of the reaction.

2 A catalyst increases the rate of the reverse reaction.

3 A catalyst increases the average kinetic energy of the reacting particles.

A. 1 only

B. 1 and 2

C. 3 only

D. 1, 2 and 3

[1 mark]

Model Answers: Easy

1

The correct answer is **A** because:

- All of the three options listed will change the rate of the reaction.
- The only option that will change the value of the activation energy is the presence of a catalyst.
- A catalyst offers an alternative pathway for the reaction with a lower activation energy.

B & **D** are incorrect as an increase in temperature will increase the number of particles with the activation energy but not the activation energy value itself.

C & **D** are incorrect as an increase in the concentration of the reactants will increase the frequency of successful collisions but not change the value of the activation energy.

2

The correct answer is **B** because:

- The Pd^{2+} ions are acting as a catalyst to the reaction.
- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.
- A catalyst does not increase the energy of the reacting molecules, so the correct option is **B.**

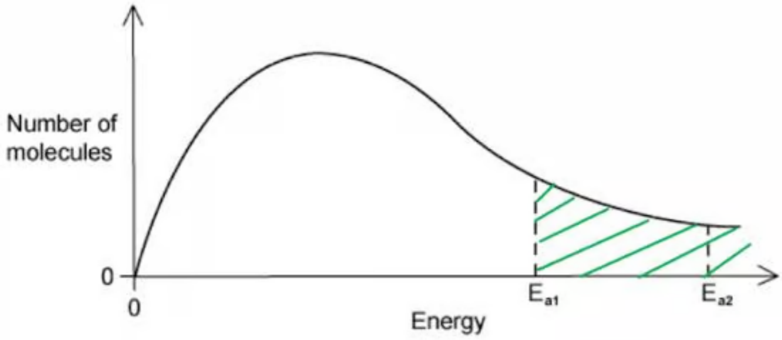
All the other statements are correct about a catalyst.

3

The correct answer is **D** because:

- The definition of a catalyst is that it **lowers the activation energy** by providing an **alternative pathway**; it does not form part of the products and is therefore not used in the reaction.

- The area under a Maxwell-Boltzmann curve is the number of particles with the activation energy.



- If a catalyst lowers the activation energy, then the number of particles with the activation energy increases.

4

The correct answer is **B** because:

- The forward reaction is exothermic as shown by the ΔH value of -93 kJ mol^{-1}, (remember a negative value is an exothermic reaction).

- If the temperature is increased the backwards reaction is favoured as the backward reaction is endothermic, this is the basis of Le Chatelier's principle.
  - If a dynamic equilibrium is disturbed by changing the conditions, the position of equilibrium moves to counteract the change.

- However, raising the temperature increases the rate of the forward AND backward reaction, because it increases the collision frequency of the particles.

- The backward reaction is momentarily favoured (so its rate is faster than the forward reaction) until a new equilibrium position is reached at which point the rates are the same.

5

The correct answer is **B** because:

- In order for a reaction to take place, the particles need to collide in the correct orientation with the right amount of energy.
- The higher the temperature, the greater the number of particles with sufficient energy to react.
- The more particles with higher energy, the greater the chance of that collision being between two particles with sufficient energy and orientation and therefore being effective.

A is incorrect as statement 2 is also correct.

C & **D** are incorrect as the activation energy of the reaction does not change; it is the number of particles with the amount of activation energy that changes.

6

The correct answer is **C** because:

- In order for a reaction to take place, the particles need to collide in the correct orientation with the right amount of energy.
- The higher the pressure, the closer the particles are and more likely to collide more frequently.

A, B & **D** are incorrect as increasing the pressure does not lower the activation energy (this is achieved with a catalyst). Increasing the pressure does not increase the energy of the particles (this is achieved with increasing the temperature).

7

The correct answer is **D** because:

- In order for a reaction to take place, the particles need to collide in the correct orientation with the right amount of energy.
- The higher the temperature, the more energy individual particles have so more particles have an energy greater than the activation energy and more likely to collide successfully.

frequently.

A, B & **D** are incorrect as increasing the pressure does not lower the activation energy (this is achieved with a catalyst). Increasing the pressure does not increase the energy of the particles (this is achieved with increasing the temperature).

7

The correct answer is **D** because:

- In order for a reaction to take place, the particles need to collide in the correct orientation with the right amount of energy.
- The higher the temperature, the more energy individual particles have so more particles have an energy greater than the activation energy and more likely to collide successfully.

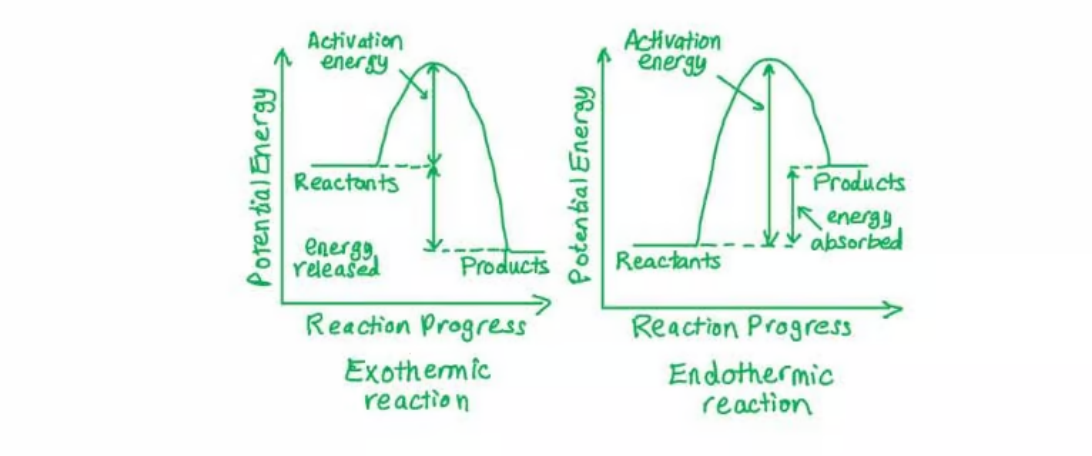
A & **B** are incorrect as the activation energy of the reaction is not changed by a change in temperature.

C is incorrect as he number of particles with energy higher than the activation energy has a much more significant effect than the kinetic energy of the particles increasing.

8

The correct answer is **D** because:

- The forward reaction is exothermic, and the backwards reaction is endothermic.
- An exothermic reaction has a negative ΔH, and an endothermic reaction has a positive ΔH.



- This reaction will have larger activation energy and a positive ΔH.

9

The correct answer is **C** because:

- In order for a reaction to take place, the particles need to collide in the correct orientation with the right amount of energy.
- The higher the temperature, the greater the number of particles with sufficient energy to react; **the rate of reaction increases.**
- The more particles with higher energy, the greater the chance of that collision being between two particles with sufficient energy and orientation and therefore effective.
- For many reactions that happen around room temperature:
  - Increasing the temperature by 10˚C doubles the rate of reaction as the number of particles with high enough energy increases.

A, B and D are incorrect as the number of particles with energy higher than the activation energy has a much more significant effect than the kinetic energy of the particles increasing.

10

The correct answer is **B** because:

- The definition of a catalyst is that it **lowers the activation energy** by providing an